

Model A Overview

 **Note:** The most accurate information regarding the model can be found in the [URDF files under the models directory](#) of the Rainbow Robotics SDK repository. It is highly recommended to refer to this source for the latest updates and details.

Model Name

A

Degrees of Freedom (DOF)

- Total DOF: 24
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Joint Names

The joint names of the A model are as follows:

- **Wheels:** `right_wheel` , `left_wheel`
- **Torso:** `torso_0` , `torso_1` , `torso_2` , `torso_3` , `torso_4` , `torso_5`
- **Right Arm:** `right_arm_0` , `right_arm_1` , `right_arm_2` , `right_arm_3` , `right_arm_4` , `right_arm_5` , `right_arm_6`
- **Left Arm:** `left_arm_0` , `left_arm_1` , `left_arm_2` , `left_arm_3` , `left_arm_4` , `left_arm_5` , `left_arm_6`
- **Head:** `head_0` , `head_1`

Mobility Indexes

The indexes related to mobility are as follows:

- **Right Wheel Index:** 0
 - **Left Wheel Index:** 1
-

Body Indexes

The indexes related to the body parts are as follows:

- **Torso Indexes:** 2, 3, 4, 5, 6, 7
 - **Right Arm Indexes:** 8, 9, 10, 11, 12, 13, 14
 - **Left Arm Indexes:** 15, 16, 17, 18, 19, 20, 21
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Head Indexes

The indexes related to the head parts are as follows:

- **Head Indexes:** 22, 23
-

Brake and Motor Information

Components Without Brakes

The following components do not have brakes:

- **Wheels:** `right_wheel` , `left_wheel`
- **Head:** `head_0` , `head_1`
- **Right Arm:** `right_arm_6`
- **Left Arm:** `left_arm_6`

Components Without Physical Brake Buttons

The following components have brakes but do not have physical buttons for releasing them (can only be released electronically):

- **Torso:** `torso_3` , `torso_4` , `torso_5`
- **Right Arm:** `right_arm_4` , `right_arm_5`
- **Left Arm:** `left_arm_4` , `left_arm_5`

Components With Physical Brake Buttons

The following components have brakes and include physical buttons for releasing them:

- **Torso:** `torso_0` , `torso_1` , `torso_2`
- **Right Arm:** `right_arm_0` , `right_arm_1` , `right_arm_2` , `right_arm_3`
- **Left Arm:** `left_arm_0` , `left_arm_1` , `left_arm_2` , `left_arm_3`

Motor Types

The motor types for each component are as follows:

- **Dynamixel Motors:** Used in the Head (`head_0` , `head_1`)
- **RBMotors:** Used in all other components (Torso, Wheels, Right Arm, Left Arm)

Parameter Lists

Parameter Name	Default Value	Min Value	Max Value
<code>robot_model_name</code>	"rby1a"	-	-
<code>model_name</code>	""	-	-
<code>battery_config</code> (<i>type, cutoff_voltage, fully_charged_voltage, ...</i>)	[0, 44.7, 58.6, 1123.0, -61.587, 1.0168, -0.0045]	-	-
<code>soft_stop_enabled</code>	0	0	1

Parameter Name	Default Value	Min Value	Max Value
<code>power_command.timeout</code>	1.0	0.0	10.0
<code>servo_on_command.timeout</code>	15.0	0.0	30.0
<code>servo_on_command.new_data_timeout</code>	3.0	0.0	10.0
<code>gravity</code>	[0, 0, -9.8]	-	-
<code>default.velocity_limit_scaling</code>	1.0	0.01	1.5
<code>default.acceleration_limit_scaling</code>	1.0	0.01	1.5
<code>default.linear_acceleration_limit</code>	5.0	0.01	10.0
<code>default.angular_acceleration_limit</code>	1.0	0.01	5.0
<code>default.optimal_control_command.velocity_tracking_gain</code>	0.1	1e-5	1.0
<code>default.optimal_control_command.stop_cost</code>	1e-3	1e-6	1e9
<code>default.se2_velocity_command.acceleration_limit</code>	[1.0, 5.0, 99.0]	-	-
<code>jog_command.one_step_size</code>	0.087266463 (5°)	0.017453293 (1°)	0.17453293 (10°)
<code>jog_command.canceling_time</code>	0.1	1e-3	10.0
<code>joint_position_command.cutoff_frequency</code>	15.0	1e-3	100.0
<code>joint_position_command.canceling_time</code>	0.1	1e-3	1.0
<code>joint_position_command.torque_saturation_time</code>	0.1	1e-3	1.0
<code>cartesian_command.cutoff_frequency</code>	15.0	1e-3	100.0
<code>cartesian_command.canceling_time</code>	0.1	1e-3	1.0
<code>cartesian_command.torque_saturation_time</code>	0.1	1e-3	1.0
<code>impedance_control_command.torque_saturation_time</code>	0.1	1e-3	1.0
<code>optimal_control_command.torque_saturation_time</code>	0.1	1e-3	1.0
<code>stop_command.canceling_time</code>	0.1	1e-3	1.0
<code>stop_command.torque_saturation_time</code>	0.1	1e-3	1.0
<code>joint_velocity_command.canceling_time</code>	0.1	1e-3	1.0

Parameter Name	Default Value	Min Value	Max Value
joint_velocity_command.torque_saturation_time	0.1	1e-3	1.0
se2_velocity_command.canceling_time	0.1	1e-3	1.0

Example Usage in Python

python

```
# Set parameters using the dot notation, following the provided examples
robot.set_parameter("default.acceleration_limit_scaling", "0.8")
robot.set_parameter("joint_position_command.cutoff_frequency", "5")
robot.set_parameter("cartesian_command.cutoff_frequency", "5")
robot.set_parameter("default.linear_acceleration_limit", "5")
```

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